

Section - I: Multiple Choice Questions

1. Which of the following scenarios best describes overfitting in a machine learning model?
 - A) The model performs well on the training data but poorly on new, unseen data.
 - B) The model performs equally well on both the training data and new, unseen data.
 - C) The model performs poorly on both the training data and new, unseen data.
 - D) The model is unable to learn from the training data and makes random predictions.

Section - II: Descriptive / Case Study-Based Questions

1. A marketing team at an e-commerce company wants to develop a machine learning model to predict customer churn. They have collected a dataset of 10,000 customers with features like demographics, purchase history, and website interactions. After training a neural network model on 80% of the data, they achieved an accuracy of 95% on the training set but only 60% on the test set. What might be the reason for this discrepancy, and what steps can the team take to address this issue?